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Impact of rapid nucleic acid amplification tests for tuberculosis on patient outcomes (Review)

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Impact of rapid nucleic acid amplification tests for tuberculosis on patient outcomes.
Cochrane Database of Systematic Reviews 2025, Issue 12. Art. No.: CD016194.
DOI: 10.1002/14651858.CD016194.

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[Intervention Review]

Impact of rapid nucleic acid amplification tests for tuberculosis on patient outcomes

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Editorial group: Cochrane Central Editorial Service.
Publication status and date: New, published in Issue 12, 2025.

Citation: Otieno JA, Were LMalesi, Lutje V, Scandrett K, Takwoingi Y, Ochodo EA. Impact of rapid nucleic acid amplification tests for tuberculosis on patient outcomes. *Cochrane Database of Systematic Reviews* 2025, Issue 12. Art. No.: CD016194. DOI: 10.1002/14651858.CD016194.

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ABSTRACT

Rationale

The World Health Organization (WHO) recommends rapid nucleic acid amplification tests (NAATs), such as Xpert MTB/RIF Ultra, as the initial diagnostic test for tuberculosis (TB). However, the effect of these tests on key health outcomes is not well-established.

Objectives

To assess the effects of three classes of rapid NAATs—low-complexity automated (LC-aNAATs), low-complexity manual (LC-mNAATs) and moderate-complexity automated (MC-aNAATs)—compared with smear microscopy or bacterial culture on patient outcomes in people investigated for TB, irrespective of age or disease severity. Our secondary objective was to assess the effects of the three classes of rapid NAATs in vulnerable populations such as children and people living with HIV, to inform health equity considerations.

Search methods

We searched Cochrane CENTRAL, MEDLINE, Embase, five other electronic bibliographic databases, and two trial registries, from 1900 up to 21 January 2025. We also checked reference lists of included articles and relevant systematic reviews to identify additional studies. We contacted manufacturers, researchers, and experts working on new diagnostic tests for TB through a WHO public call for data, which lasted from 30 November 2023 to 15 February 2024.

Eligibility criteria

We included randomised controlled trials (RCTs) that involved children and adults of any age who were suspected to have TB (i.e. 'presumptive TB'). We included both individually randomised and cluster-randomised designs. In terms of the intervention, we considered "design-locked, marketed test technologies" (i.e. testing technologies with designs that have been finalised and are currently being sold or promoted in the healthcare market) by class and tests provided by the WHO: LC-aNAATs (e.g. Xpert MTB/RIF Ultra, Truenat MTB Plus, Truenat MTB-RIF Dx); LC-mNAATs (e.g. TB-LAMP); and MC-aNAATs (e.g. FluoroType XDR-TB). We restricted comparative strategies to smear microscopy and bacterial culture methods. We excluded studies on standard Xpert MTB/RIF, which has been superseded by the Xpert MTB/RIF Ultra test.

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Outcomes

Our critical outcomes were all-cause mortality from the time of the first diagnostic test to the end of the trial follow-up, and cure rates among people with bacteriologically-confirmed TB at the beginning of treatment and who completed treatment with no evidence of failure. Our important outcomes were the proportion of participants lost to follow-up, time to TB diagnosis, and time to treatment initiation.

Risk of bias

Two review authors independently assessed the risk of bias for all reported outcomes in the included studies using the revised Cochrane risk of bias tool (RoB 2). Disagreements were resolved through discussion or by consulting a senior review author as needed.

Synthesis methods

We could not combine the results statistically because the included studies measured, defined, or reported outcomes differently. As a result, we presented the findings from each study in tables, forest plots, and using narrative descriptions. We used GRADE to assess the certainty of evidence for each outcome using GRADEpro GDT software.

Included studies

Of the 2231 records identified by our search, we included two trials (reported in three articles) in the review. The trials compared an LC-aNAAT (Xpert MTB/RIF Ultra) with smear microscopy and involved 11,228 adults aged 18 years and above. One was an individually randomised trial and one was cluster-randomised. Both trials were conducted in Africa (one in South Africa and the other in Uganda). One trial assessed all-cause mortality at six and 18 months, loss-to-follow-up, and the time taken to diagnose TB, while the other assessed the time taken to initiate treatment.

Synthesis of results

Whether Xpert MTB/RIF Ultra or smear microscopy was used to detect TB made little to no difference to all-cause mortality at six months (risk ratio 0.79, 95% CI 0.61 to 1.01; 1 RCT, 8116 participants; high-certainty evidence) and 18 months (adjusted rate ratio 0.77, 95% confidence interval (CI) 0.46 to 1.29; 1 RCT, 8413 participants; high-certainty evidence).

Neither study assessed how the test used affects the likelihood of curing TB.

One trial (XPEL-TB 2021) reported a slightly higher risk of loss to follow-up with Xpert MTB/RIF Ultra (13.6%) than with smear microscopy (10.1%), with a risk ratio of 1.35 (95% CI 1.20 to 1.51; 1 RCT, 9563 participants; high-certainty evidence).

A testing strategy that includes Xpert MTB/RIF Ultra probably results in a reduction in time to diagnosis (adjusted geometric mean ratio 0.49, 95% CI 0.39 to 0.62; 1 RCT, 10,644 participants; moderate-certainty evidence), and a similar reduction in time to treatment (adjusted geometric mean ratio 0.35, 95% CI 0.21 to 0.58; 1 RCT, 10,644 participants; hazard ratio 3.00, 95% CI 1.18 to 7.60; 1 RCT, 584 participants; moderate-certainty evidence).

The evidence in this review is limited by sparse data, a lack of some key outcomes (specifically, cure), the use of varied effect measures, and the exclusion of paediatric populations. While we graded the evidence as moderate certainty for time to diagnosis and treatment initiation, serious concerns about the risk of bias and the narrow scope restrict the applicability and generalisability of the evidence to other NAATs and patient outcomes.

Authors' conclusions

Based on limited evidence from two trials that compared Xpert MTB/RIF Ultra to smear microscopy, the use of Xpert MTB/RIF Ultra to test for TB makes little to no difference to all-cause mortality at six and 18 months. Xpert MTB/RIF Ultra probably decreases the time taken to diagnose TB and time to treatment initiation. The use of Xpert MTB/RIF Ultra increases the risk of loss to follow-up compared with smear microscopy. There were no data on the effects of Xpert MTB/RIF Ultra compared with smear microscopy or culture on tuberculosis cure rates.

Funding

This Cochrane review was funded by the World Health Organization (WHO).

Registration

The protocol is available via the Open Science Framework registry at https://osf.io/puax2.

PLAIN LANGUAGE SUMMARY

Do rapid genetic tests (molecular tests) for tuberculosis improve health outcomes compared to tests traditionally used to detect TB (microscopy and culture)?

Key messages

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- Xpert MTB/RIF Ultra is a rapid test that probably reduces the time taken to diagnose tuberculosis (TB) and the time from diagnosis to starting treatment.
- Using Xpert MTB/RIF Ultra does not reduce deaths from any cause occurring within 18 months.
- The studies in this review did not measure whether rapid tests improve recovery from TB (cure).

What is tuberculosis?

Tuberculosis (TB) is a serious bacterial infection that primarily affects the lungs. TB causes symptoms such as cough, fever, and weight loss. It can also spread to other parts of the body. Children, especially under-fives, are more vulnerable and may develop mild to severe forms of the disease. TB affects many people each year, and is largely curable, though there is a risk of death if standard treatments do not work. Drug-resistant TB is a type of TB that is harder to treat because it does not respond to common medicines.

Why are rapid tests being used?

Traditional diagnostic methods for TB, such as smear microscopy and culture, use the examination of samples under a microscope (an optical instrument that uses a lens to magnify the bacteria) or grow bacteria in a laboratory. A newer type of test known as a molecular test (or 'nucleic acid amplification test') magnifies genetic material in a sample of bodily fluid (e.g. phelgm (saliva)) many times to detect if TB bacteria are present. Some of the molecular tests, such as Xpert MTB/RIF Ultra, can also detect resistance to rifampicin, which is a widely used TB medicine. The molecular tests have been proposed to detect TB more quickly and accurately. This may help patients receive appropriate treatment more quickly.

There are various types of rapid molecular TB tests, which differ in how they are performed and where they are used. Some are conducted in laboratories, whereas others are used in clinics or at the point of care. The World Health Organization (WHO) has categorised the tests into three classes, which they have called laboratory-based automated tests (e.g. Xpert MTB/RIF Ultra, Truenat MTB-RIF Dx, and Truenat MTB Plus), manual laboratory tests (e.g. TB-LAMP), and moderately complex automated tests (e.g. FluoroType XDR-TB). Although health workers generally decide which test to use, patients in private clinics may be able to choose from the test options available.

What did we want to find out?

We wanted to find out whether rapid TB tests affect health outcomes, including death, recovery (cure), discontinuation of treatment, and time taken to diagnose TB and to start treatment.

What did we do?

We searched for evidence from studies known as 'clinical trials' that compared rapid TB tests with traditional methods of testing for TB, like smear microscopy and culture. We summarised the findings and made a judgement about the reliability of the results, based on how the studies were designed and the number of people who took part.

What were the main results of the review?

We identified two clinical trials from three reports published between 2021 and 2024. The two trials involved 11,228 adults being tested for TB. Most participants were from Uganda (a low-income country), and a smaller number were from South Africa (an upper-middle-income country). No information was available on cure rates.

Although our review focused on rapid TB tests in general, all available evidence came from two trials that both evaluated the Xpert MTB/RIF Ultra test. This test did not reduce deaths from any cause at six months or18 months. However, it probably sped up the process of diagnosing TB and starting treatment when compared to traditional testing methods. People tested with Xpert MTB/RIF Ultra were more likely to stop treatment than those tested with smear microscopy. The studies did not measure whether the rapid TB tests improved recovery.

What were the limitations of the evidence?

We found evidence only for the Xpert MTB/RIF Ultra test and not for any other rapid TB test. None of the studies reported how these tests affected patient recovery. All the included studies focused on adults, so it is unclear whether the findings apply to children and adolescents. The studies also used different statistical methods to report time to diagnosis and treatment, making it difficult to compare their results. We rated our confidence in the evidence as moderate because the largest trial did not clearly describe how it measured or planned to analyse the time taken to diagnose TB and the time taken to start treatment.

How up to date is this evidence?

The evidence is based on searches of medical literature run up to 21 January 2025.

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SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Xpert MTB/RIF Ultra compared to smear microscopy for adults aged ≥ 18 years with presumptive TB

Xpert MTB/RIF Ultra compared to smear microscopy for adults aged ≥ 18 years with presumptive TB

Patient or population: adults aged ≥ 18 years with presumptive TB
Setting: healthcare facilities in Uganda and South Africa
Intervention: Xpert MTB/RIF Ultra
Comparison: smear microscopy

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with smear microscopy	Risk with Xpert MTB/RIF Ultra				
All-cause mortality at 18 months	4 per 100	**3 per 100** (5 to 2)	**Rate ratio 0.77** (0.46 to 1.29)	8413 (1 RCT)	⊕⊕⊕⊕ High[a]	Xpert MTB/RIF Ultra results in little to no difference in all-cause mortality at 18 months.
All-cause mortality at six months	3 per 100	**3 per 100** (3 to 2)	**RR 0.79** (0.61 to 1.01)	8116 (1 RCT)	⊕⊕⊕⊕ High[a]	Xpert MTB/RIF Ultra results in little to no difference in all-cause mortality at six months.
Cure - not reported	-	-	-	-	-	No study reported this outcome
Proportion of participants lost-to-follow-up	10 per 100	**14 per 100** (15 to 12)	**RR 1.35** (1.20 to 1.51)	9563 (1 RCT)	⊕⊕⊕⊕ High	Xpert MTB/RIF Ultra results in a slight increase in proportion of participants lost-to-follow-up.
Time to diagnosis	The adjusted geometric mean ratio for time to diagnosis (1 trial, 10,644 participants) was 0.49 (95% CI 0.39-0.62).			10644 (1 RCT)	⊕⊕⊕⊖ Moderate[b]	Xpert MTB/RIF Ultra probably reduces the time to diagnosis.
Time to treatment	The adjusted geometric mean ratio for time to treatment (1 trial, 10,644 participants) was 0.35 (95% CI 0.21-0.58). Hazard ratio for time to treatment (1 trial, 584 participants) was 3.00 (95% CI 1.18-7.60) for estimates without culture diagnoses.			11228 (2 RCTs)	⊕⊕⊕⊖ Moderate[b]	Xpert MTB/RIF Ultra probably reduces time to treatment.

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*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the __relative effect__ of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_452675698239626394.

[a] Not downgraded for imprecision because the total sample size met the optimal information size and the confidence interval excludes appreciable benefit or harm, despite spanning no effect. Mortality is objective and at low risk of bias.
[b] Downgraded 1 level for risk of bias: some concerns about risk of bias were identified for Cattamanchi 2021, which was the largest trial included in this review, because key details on the measurement of time to treatment and diagnosis by study arm were lacking. The protocol and statistical analysis plan also did not clearly pre-specify outcome measures or analytic strategies, for the time to diagnosis and treatment outcomes hence, limiting confidence in the reported results.

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BACKGROUND

Description of the condition

Tuberculosis (TB) is a leading cause of death worldwide, affecting individuals of all ages [1]. The bacterium *Mycobacterium tuberculosis* causes TB disease. While TB is preventable and largely curable, an estimated 10.8 million people became ill with TB globally in 2023, and approximately 1.25 million people died from the disease [2, 3]. TB primarily impacts the respiratory system, causing pulmonary TB (PTB), which is characterised by loss of appetite, weight loss, lethargy, fever, chills, night sweats, cough, and haemoptysis (coughing up blood). It can also affect other organ systems in the form of extrapulmonary TB (EPTB), which can impact every part of the body, with lymph nodes and the pleura being the most common sites [4]. EPTB is noninfectious and can occur alone or in combination with PTB, causing signs and symptoms related to the affected structures.

The natural history of TB in children differs from that of adults. Children and adolescents are more likely to progress from initial infection with *Mycobacterium tuberculosis* to primary TB disease, with those under five years being particularly susceptible [5]. The clinical spectrum in children ranges from asymptomatic primary infection to symptomatic pulmonary disease, extrapulmonary involvement, and in some older children and adolescents, “adult-type” disease, characterised by cavitation and higher bacillary burden [6].

TB disproportionately affects vulnerable populations, including young children, people living with HIV, and those in resource-limited settings. These groups often face structural barriers to timely diagnosis, access to treatment, and continuity of care, which can lead to poorer health outcomes. For example, children under five years of age have been shown to have higher TB case fatality compared to older children [7], while people living with HIV or those who develop drug-resistant TB may experience delays in diagnosis or receive suboptimal treatment [8].

Among these vulnerable populations, the emergence and spread of drug-resistant TB pose an additional challenge, as timely detection and appropriate treatment are critical to improving outcomes. By contrast, drug-susceptible tuberculosis (DS-TB) refers to TB caused by strains of *Mycobacterium tuberculosis* that remain susceptible to standard first-line anti-TB drugs, such as rifampicin and isoniazid. Drug-resistant TB (DR-TB) is a global health threat, affecting hundreds of thousands annually, and is responsible for a substantial proportion of deaths associated with antimicrobial resistance [3, 9]. Detecting drug resistance requires bacteriological confirmation of TB and testing using rapid molecular tests, culture methods, or sequencing technologies. Resistance to rifampicin, the key first-line drug for TB, is particularly important because it alone signals a high likelihood of multidrug resistance, and when it co-occurs with isoniazid resistance, it defines multidrug-resistant TB (MDR-TB), which requires treatment with second-line drugs. [2]. The World Health Organisation (WHO) classifies DR-TB cases into five categories: isoniazid-resistant TB, rifampicin-resistant TB, MDR-TB, pre-extensively drug resistant TB (pre-XDR-TB), and XDR-TB [10]. Pre-XDR-TB is caused by strains that are multidrug-resistant (MDR/RR-TB) and resistant to any fluoroquinolone, while XDR-TB is caused by MDR/RR-TB strains that are resistant to fluoroquinolones and at least one additional Group A drug, which includes levofloxacin or moxifloxacin, bedaquiline, and linezolid [10].

Conventional diagnostic methods for tuberculosis include smear microscopy and culture [3]. Smear microscopy is widely used in many settings because it is inexpensive and straightforward, but it has lower sensitivity, especially in children and people living with HIV. Culture is considered the reference standard due to its higher sensitivity, but it is slow and requires specialised laboratory infrastructure. Globally, in 2023, only around 62% of people diagnosed with pulmonary TB were bacteriologically confirmed using these methods or WHO-recommended rapid tests, indicating that much diagnostic testing still relies on traditional approaches [3].

Description of the intervention and how it might work

WHO now recommends several rapid molecular tests as the initial diagnostic test for TB, some of which can simultaneously detect drug resistance [11]. Some of these tests can be used in any part of the health system, while others, such as those of moderate complexity, require facilities with adequate laboratory capacity. Due to the increasing number of rapid molecular diagnostic tools for TB, many of which detect TB and drug resistance using similar underlying technologies, the WHO introduced a class-based approach in December 2020, with a Guideline Development Group (GDG) recommending three new test classes. WHO groups individual tests with similar characteristics and performances into a class. The classes are defined by the type of technology (e.g. automated or reverse hybridisation nucleic acid amplification tests (NAATs)), the complexity of the implementation test (e.g. low, moderate, or high—considering the requirements of infrastructure, equipment, and technical skills of laboratory staff), and the target conditions (e.g. diagnosis of TB and detection of resistance to first-line or second-line drugs) [12]. The three classes are: low-complexity automated NAATs (LC-aNAATs), low-complexity manual NAATs (LC-mNAATs), and moderate-complexity automated NAATs (MC-aNAATs).

For the three classes, we describe the 'design-locked, marketed test technologies' (i.e. testing technologies whose designs have been finalised and are currently being sold or promoted in the market) based on information from the WHO Global TB Programme and other publicly available sources [13]. For LC-aNAATs, we considered Xpert MTB/RIF Ultra, Truenat MTB Plus, Truenat MTB-RIF Dx, STANDARD M10, and IRON-qPCR (quantitative polymerase chain reaction). Xpert MTB/RIF Ultra diagnoses TB and RR-TB and can be performed by operators with minimal technical expertise. Additionally, Xpert MTB/RIF Ultra demonstrates high diagnostic accuracy, with sensitivities of 90.9% for PTB and 94.9% for RR-TB, and specificities of 95.6% for PTB and 99.1% for RR-TB, according to evidence from Cochrane reviews [14, 15]. While Truenat MTB Plus provides semi-quantitative results with 80% sensitivity for detecting pulmonary TB, Truenat MTB-RIF Dx specifically targets the *rpoB* gene to detect rifampicin resistance [16, 17, 18]. The Standard M10 is an in vitro molecular diagnostic system that combines isothermal amplification and real-time PCR on a single platform, with two TB variants, Standard M10 MTB/NTM and Standard M10 MDR-TB, that deliver qualitative results within 80 minutes from sputum specimens [19]. Iron-qPCR is a point-of-care molecular diagnostics system that combines nucleic acid extraction and real-time PCR for syndromic diagnostics (simultaneous detection of multiple pathogens or conditions),

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allowing the detection of over 10 gene targets (specific DNA or RNA sequences used to identify pathogens or resistance markers) in a single cartridge test with a multiplexing capacity of up to 40 targets (number of gene targets that can be detected simultaneously), providing results in 40 minutes [20].

For the LC-mNAATs, this review focused on TB loop-mediated isothermal amplification (TB-LAMP). This rapid molecular method amplifies DNA at a constant temperature using a set of specially designed primers, enabling detection without sophisticated thermal-cycling equipment. TB-LAMP is a low-cost test that serves as an alternative to smear microscopy for diagnosing TB in resource-limited settings and requires similar biosafety measures and infrastructure [21, 22, 23]. While it has shown higher sensitivity (94.9%) and specificity (89.8%) than smear microscopy in both children and adults, TB-LAMP does not provide information on rifampicin resistance or resistance to other TB drugs [24, 25].

For MC-aNAATs, this review focused on the qualitative, in vitro FluoroType XDR test, which automates the detection of the *M. tuberculosis* complex and its resistance to fluoroquinolones, amikacin (an aminoglycoside antibiotic, used as a second-line drug), ethambutol (a first-line anti-TB drug), and linezolid (a second-line oxazolidinone antibiotic) from sputum and cultivated samples [26]. Resistance is identified by detecting specific mutations in the *gyrA* and *gyrB* genes for fluoroquinolones, *rrs* for amikacin, and *embB*, *rplC*, and *rrl* for ethambutol and linezolid, respectively [26, 27, 28]. In this review, we referred to the FluoroType XDR-TB test, which builds on the earlier FluoroType MTBDR test by detecting resistance to additional second-line drugs.

Rapid NAATs have higher diagnostic accuracy than smear microscopy, can detect drug resistance, can be deployed at the point of care, and have short turnaround times, which may positively impact patient-important outcomes [29]. We considered several hypotheses that follow from this. The use of these tests may result in a decrease in pre-diagnostic and pre-treatment loss to follow-up and, perhaps, reduce time to treatment [29]. Pre-treatment loss to follow-up is associated with high mortality; therefore, reducing it may lower mortality [30]. A larger proportion of people with true TB may receive effective therapy, and fewer individuals may be falsely diagnosed with the disease on clinical grounds alone or incorrectly treated [29]. Rapid detection of drug resistance may enable earlier initiation of effective treatment. These factors could potentially improve important patient outcomes [29].

Why it is important to do this review

A Cochrane review assessed the impact of diagnostic strategies using the standard Xpert MTB/RIF test compared with smear microscopy on patient-important outcomes [29]. The evidence from that review informed the 2021 update of the WHO policy guidelines on TB detection tests [31]. However, since that time, Xpert MTB/RIF has been superseded by Xpert MTB/RIF Ultra, and other rapid NAATs have become available. Generating evidence that reflects the latest technological advancements is helpful for guideline development groups and policymakers in decision-making. In preparation for the Guideline Development Group (GDG) meeting "Nucleic Acid Amplification Tests to Detect TB and DR-TB," held from 6 to 10 May 2024, our team summarised evidence on the impact of LC-aNAATs for the detection of TB and drug resistance, LC-mNAATs for the detection of TB alone, and MC-aNAATs for the detection of pre-XDR-TB or XDR-TB on patient outcomes for TB and drug resistance [32]. Our review assessed the impact of these tests on patient-important outcomes, including all-cause mortality, cure, proportion lost to follow-up, time to confirmed diagnosis, and time to treatment initiation for TB.

Given the previously mentioned disparities, such as higher TB case fatality ratios amongst young children, increased vulnerability amongst people living with HIV, and geographic differences in healthcare access and TB burden, this review incorporated an equity perspective by assessing intervention effects across key subgroups, including age, HIV status, drug resistance, and geographical region. Understanding whether the effects of interventions differ across these groups is essential for informing equitable policy and program design, especially given the wide variation in TB burden and health system access across settings.

OBJECTIVES

To assess the effects of three classes of rapid NAATs—low complexity automated (LC-aNAATs), low complexity manual (LC-mNAATs) and moderate complexity automated (MC-aNAATs)—compared with smear microscopy or bacterial culture on patient outcomes in people investigated for TB irrespective of age or disease severity.

Our secondary objective was to assess the effects of the three classes of rapid NAATs in vulnerable populations such as children and people living with HIV, to inform health equity considerations.

METHODS

We followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR), the *Cochrane Handbook for Systematic Reviews of Interventions*, and Cochrane plain-language summary guidance when conducting the review, and PRISMA 2020 guidelines when reporting the process and results [33, 34, 35, 36].

Differences between protocol and review

We were unable to implement the following planned methods due to insufficient or no data.

- Dealing with missing data: planned sensitivity analyses to assess the impact of missing data could not be performed.
- Synthesis methods: random-effects meta-analysis was prespecified but could not be conducted because the included studies were too heterogeneous.
- Investigation of heterogeneity and subgroup analyses: planned analyses by age, HIV status, drug resistance, and WHO region were not feasible.
- Reporting bias assessment: funnel plot assessment was not applicable due to the small number of studies.
- Sensitivity analyses: these could not be conducted for missing data, outliers, high risk of bias, or manufacturer-supplied data.
- Equity-related assessment: although we collected data on vulnerable populations (e.g. children versus adults), a separate synthesis was not feasible due to limited data.
- Consumer involvement: direct engagement was not possible due to resource constraints, though we used core outcome sets developed with consumer input and issued a public call for unpublished data.

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We made it clear that observational studies with before-and-after implementation designs were not eligible for inclusion. Although this exclusion was always intended, it had not been explicitly stated in the protocol [37].

Criteria for considering studies for this review

Types of studies

We included cluster-randomised controlled trials and individually randomised controlled trials that met our inclusion criteria. We excluded observational studies with before-and-after implementation periods, as these designs are more susceptible to bias from secular trends and confounding in the absence of concurrent control groups. Cross-over trials were ineligible for inclusion because the intervention under study (diagnostic testing strategy) results in irreversible clinical management and outcomes (e.g. treatment initiation, time to diagnosis, mortality). As such, our outcomes of interest could not be repeated within the same individual, making cross-over designs inappropriate.

Types of participants

We included people with presumptive TB irrespective of disease severity. We considered children separately from adults and adolescents in this review. Based on the WHO definition of age groups, we defined children as those under 10 years of age [38].

Types of interventions

We considered only diagnostic strategies involving design-locked, marketed test technologies categorised by class. LC-aNAATs included Xpert MTB/RIF Ultra, Truenat MTB Plus, Truenat MTB-RIF Dx, STANDARD M10, SD Biosensor, IRON-qPCR and BIONEER; LC-mNAATs included TB-LAMP; and MC-aNAATs included FluoroType XDR-TB. We considered diagnostic strategies that used smear microscopy and/or culture as the controls. Traditionally, TB diagnosis has relied on smear microscopy and culture methods; however, culture-based methods are considered the reference standard for confirming TB due to their higher sensitivity and ability to detect viable *Mycobacterium tuberculosis* [39]. Smear microscopy, while still used in many resource-limited settings, has limited sensitivity, particularly in people living with HIV and in children. According to WHO guidelines, rapid molecular diagnostics, such as Xpert MTB/RIF Ultra and other NAATs, are recommended as initial diagnostic tests for all individuals with signs and symptoms of TB [31]. We excluded studies of the standard Xpert MTB/RIF test because it has been superseded by the Xpert MTB/RIF Ultra test [40].

Outcome measures

Critical outcomes

- All-cause mortality from the time of the first diagnostic test to the end of trial follow-up
- Cure

For all outcomes, we used the WHO definitions (see Table 1). The WHO consulted on the definitions of DR-TB treatment outcomes in 2020 [41]. Following this consultation, the WHO proposed new definitions of TB treatment outcomes applicable to the assessment of outcomes across all forms of TB [41].

Important outcomes

- Proportion of participants lost to follow-up
- Time to diagnosis
- Time to treatment

Search methods for identification of studies

We tried to identify all relevant trials, regardless of language or publication status.

Electronic searches

We initially conducted searches on 16 October 2023 and updated the searches on 21 January 2025. We searched the following databases, with the assistance of an information specialist (review author VL), using the search terms and strategy described in Supplementary material 1.

- Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 12) in the Cochrane Library (searched 21 January 2025)
- MEDLINE(R) ALL Ovid (1946 to 21 January 2025)
- Embase Ovid (1947 to 21 January 2025)
- Science Citation Index Expanded ISI Web of Science (SCI-EXPANDED; 1900 to 21 January 2025)
- BIOSIS previews Web of Science (1926 to 21 January 2025)
- Scopus Elsevier (1970 to 21 January 2025)
- ProQuest Dissertations & Theses A&I (searched 21 January 2025; ProQuest | Better research, better learning, better insights)
- WHO Global Index Medicus (searched 21 January 2025; www.globalindexmedicus.net)

We used the Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE (Lefebvre 2023) and adapted it for other databases. We also searched the WHO International Clinical Trials Registry Platform (ICTRP; apps.who.int/trialsearch) and ClinicalTrials.gov (clinicaltrials.gov/ct2/home) for trials in progress, on 21 January 2025.

Searching other resources

We examined the reference lists of included articles and relevant review articles identified through the electronic searches. We searched for ongoing and unpublished studies from manufacturers and researchers through a WHO public call that was live between 30 November 2023 and 15 February 2024; and from experts working on new diagnostics for TB, such as STOP TB Partnership's New Diagnostic Working Group and FIND (a global alliance for diagnostics) [42, 43]. We also checked MEDLINE via PubMed for post-publication amendments such as expressions of concern, errata, corrigenda, or retractions, related to the included or otherwise eligible studies. We incorporated any relevant updates into our assessment.

Data collection and analysis

Review authors were not involved in eligibility decisions regarding their own studies. None of the studies eventually included in the review was authored by any of the review authors.

Selection of studies

We used Covidence systematic review management software to select studies [44]. Two review authors (JAO and LMW)

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independently screened study titles and abstracts for eligibility. Any disagreements were resolved through discussion between the review authors or, if necessary, by consulting a senior review author (EO). Afterwards, we retrieved the full-text articles of potentially eligible studies and two review authors (JAO and LMW) independently assessed their eligibility. Any disagreements were resolved through discussion or consultation with a third review author. We contacted the study authors for clarification of methods and other information, as needed.

Data extraction and management

Review authors worked with the statisticians (YT and KS) to develop a standardised data extraction form for use in Covidence [44]. Two review authors (JAO and LMW) independently piloted the form on two eligible studies. We modified the form based on the piloting outcomes, after which all review authors approved its use.

Two review authors (JAO and LMW) independently extracted data and discussed inconsistencies to achieve consensus. A senior review author (EO) was consulted where necessary. We extracted the following information: first author, year of publication, study period, funding, participant characteristics, intervention, control, each outcome measured and how it was measured, the covariates for which outcomes were adjusted, length of follow-up, and measure of effect, with standard errors or 95% confidence intervals (CIs). For cluster-RCTs, we also extracted the number of participants and clusters randomised to each diagnostic arm, the number of participants monitored for each outcome, the number of events, the intraclass correlation coefficient, and any adjusted measures of effect, along with the method of adjustment, where reported.

Risk of bias assessment in included studies

Two review authors (JAO and LMW) independently assessed the risk of bias in the included studies for each outcome listed below, using the revised Cochrane risk of bias tool (RoB 2, 22 August 2019 version) for RCTs [45, 46]. We assessed the effect of assignment to intervention (i.e. the 'intention-to-treat' effect). Any disagreements were resolved through discussion between the review authors or, if necessary, by consulting a senior review author (EO).

- All-cause mortality from the time of the first diagnostic test to the end of trial follow-up
- Cure
- Proportion of participants lost to follow-up
- Time to diagnosis
- Time to treatment

The RoB 2 tool covers five domains.

1. Bias arising from the randomisation process
2. Bias due to deviations from intended interventions
3. Bias due to missing outcome data
4. Bias in the measurement of the outcome
5. Bias in the selection of the reported result

Within each domain, we assessed the risk of bias as low, some concerns, or high, according to the RoB 2 guidance. An overall risk of bias judgement for each outcome was derived using the RoB 2 algorithm.

For the cluster-RCT we included, we applied the RoB 2 extension for cluster-RCTs, 18 March 2021 version [47]. This extension comprises three additional signalling questions for the first domain that address bias arising from the timing of participant identification and recruitment relative to cluster-randomisation. We did not apply the RoB 2 extension to cross-over trials, as these studies were ineligible [48].

The detailed risk of bias assessments, including all responses to signalling questions, are available from the review authors upon reasonable request (see Supplementary material 4).

Measures of treatment effect

For time-to-event outcomes, such as time to diagnosis or treatment, we used hazard ratios (HRs) or geometric mean ratios with 95% CIs. We planned to use risk ratio (RR) or odds ratio (OR) with their 95% CIs for the outcomes of all-cause mortality, cure, and proportion of participants lost to follow-up.

Unit of analysis issues

For cluster-randomised trials, we planned to account for clustering by using cluster-level summary measures. Where studies reported adjusted effect estimates accounting for clustering, these were used in preference to unadjusted estimates.

Dealing with missing data

We did not perform specific analyses to address missing data, as the review relied on study-level data reported in published articles. Both included studies provided only limited information on missing or excluded participants and on how missing outcome data were handled. Consistent with our protocol, we reported missing data exactly as described in the original studies.

Reporting bias assessment

We planned to evaluate small-study effects by constructing funnel plots for each meta-analysis that included at least 10 studies [45]. The minimum number of studies required to interpret funnel plot asymmetry meaningfully was not met; therefore, this analysis was not applicable.

Synthesis methods

We did not pool individual study findings statistically because of differences in how outcomes were measured, defined, or reported across the included studies. In our protocol, we prespecified that meta-analysis would be conducted only when sufficiently homogeneous trials were available, and cluster-RCTs would be adjusted using the design effect $(1 + (M - 1) \times ICC)$ before pooling with individually randomised trials [37]. Given the limited number of included studies and heterogeneity in outcome definitions, we presented individual study results narratively, using tables and forest plots.

Investigation of heterogeneity and subgroup analysis

We did not assess statistical heterogeneity using the I^2 statistic, nor did we conduct planned subgroup analyses (e.g. by age group, HIV status, drug resistance status, or WHO region), as we did not perform any meta-analyses. The two included studies differed too much in design, population characteristics, interventions, and outcome reporting to permit meaningful statistical pooling.

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Therefore, these analyses were not applicable. We presented findings narratively instead.

Equity-related assessment

We collected data on the impact of rapid TB NAATs on vulnerable populations, including individuals living with HIV, across WHO regions.

Sensitivity analysis

We did not conduct sensitivity analyses to explore the impact of missing data, outliers, high risk of bias, and manufacturer data because we did not perform any meta-analyses.

Certainty of the evidence assessment

Two independent review authors (JAO and EO) assessed the certainty of evidence using the GRADE approach and produced a 'summary of findings' table using GRADEpro GDT software [49, 50]. Disagreements were resolved by discussion among the review authors. We prioritised the following outcomes: all-cause mortality censored at six and 18 months post-diagnosis; cure; proportion of participants lost to follow-up; time to diagnosis; and time to treatment for each type of rapid NAAT, compared with smear microscopy and/or culture. We rated the certainty of the evidence for each outcome as described by Balshem and colleagues [50]. We downgraded the level of certainty in the evidence for the following main reasons: risk of bias, inconsistency, indirectness, imprecision or publication bias. The following criteria were applied to interpret the findings [50].

- High: we are very confident that the true effect lies close to that of the effect estimate.
- Moderate: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
- Very low: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of the effect.

Consumer involvement

Due to limited resources, we were unable to engage directly with consumers. However, we used core outcome sets for the review outcomes, which were developed with consumer involvement in prior related reviews. Further, WHO issued a public call to manufacturers, national TB programmes, implementers, researchers, and other agencies to submit unpublished data on NAAT technologies, including diagnostic accuracy, implementation, and economic or qualitative outcomes, to inform WHO guideline development [13]. In line with the review protocol [37], we specifically sought information on ongoing and unpublished studies, and the call was open from 30 November 2023 to 15 February 2024 [13].

RESULTS

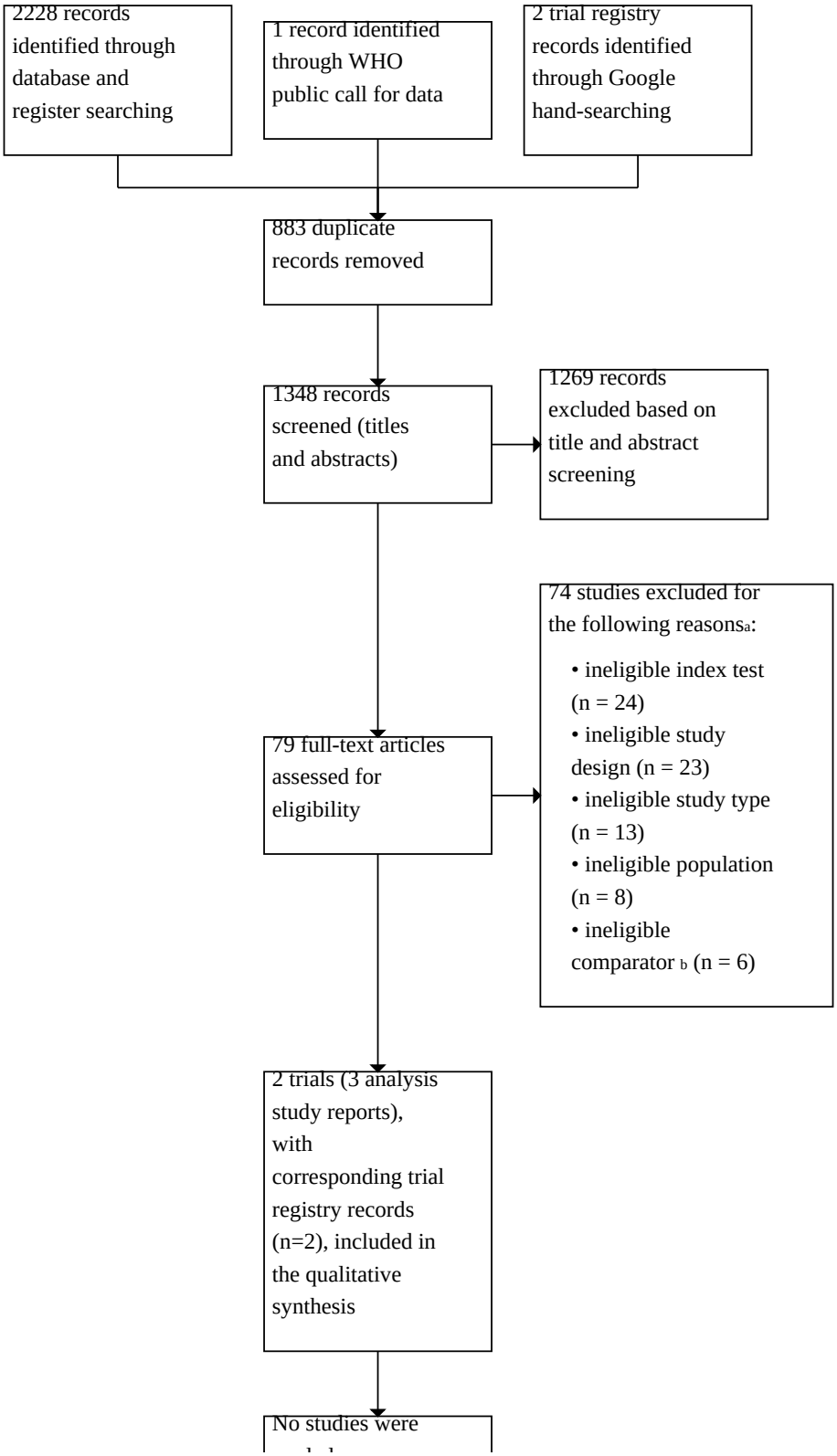
Description of studies

Results of the search

Our search yielded 2231 records, including one RCT identified through the WHO public call (Figure 1). Duplicates were removed using automated tools before screening, resulting in 883 fewer records. We screened 1348 titles and abstracts and excluded 1269 records. We retrieved and screened the full texts of the remaining 79 records against the eligibility criteria and excluded 74. We included two clinical trials (reported in three study reports), along with their corresponding trial registry records (n = 2) [51, 52, 53, 54, 55].

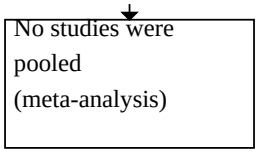
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Figure 1. Figure 1: PRISMA 2020 flow chart illustrating the study selection process [a]Reasons for exclusion are outlined for each study in Supplementary material 3. [b]Additional details are provided for three of these studies for context in Supplementary material 7.



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Figure 1. (Continued)



Included studies

We have summarised the characteristics of the included trials in Table 2 and Supplementary material 2.

Study design

We included two trials: a pragmatic cluster-randomised trial and an individually randomised trial [51, 52, 53]. Both trials were open-label, parallel-design trials using sputum samples. The XPEL-TB 2021 [52, 53, 54] trial (NCT03044158) was conducted over one year and four months using healthcare centres as the units of randomisation, whereas Aliasgar 2023 [51, 55] (NCT03168945) was performed for two years and three months [51, 52, 53]. The length of follow-up varied, with XPEL-TB 2021 reporting an 18-month follow-up period and Aliasgar 2023 having relatively short follow-up (60 days) [51, 52, 53].

Settings

Both included studies were conducted in Africa [51, 52, 53]. The XPEL-TB 2021 trial was conducted in Uganda (a low-income country) across 20 healthcare centres in urban and rural areas [52, 53]. The other trial by Aliasgar 2023 was conducted in a community-based, scalable mobile clinic in a peri-urban area in South Africa (an upper-middle-income country) [51]. The authors described this model as scalable because the mobile clinic and portable Xpert system were designed to be lower cost, easier to deploy, and more feasible to replicate in similar resource-limited settings than larger, more expensive mobile units [51].

Participants

Both trials enroled adult participants aged 18 years and above who were undergoing evaluation for presumptive TB, though the eligibility criteria varied. The XPEL-TB 2021 trial included 10,644 adults listed on the national TB and leprosy program of individuals with presumptive TB, laboratory-confirmed TB, or whose details appeared in the TB treatment registers; individuals with missing age data or unknown human immunodeficiency virus (HIV) status were excluded [52, 53]. In the secondary analysis report of the XPEL-TB 2021 trial, 9563 of the 10,644 adults were included; participants whose vital status could not be ascertained, those who were known to be alive but did not have a usable (valid) date recorded for the mortality outcome, and those with unknown HIV status were excluded from this analysis by the trial authors [53]. Aliasgar 2023 recruited 584 adults who had at least one TB symptom for two weeks or who were living with HIV, confirmed via on-site testing or self-reporting; the trial excluded HIV-negative individuals with no TB symptoms, individuals with recent or recurrent TB treatment, individuals who were unable to provide informed consent or contact details, and those recruited from household settings [51].

Interventions

Participants were randomised in a 1:1 ratio in both trials [51, 52, 53]. For the primary analysis report from the XPEL-TB 2021 trial, 5546 participants were randomised to a multicomponent diagnostic strategy (on-site Xpert MTB/RIF Ultra testing, guided restructuring of clinic workflows, and monthly feedback of quality metrics) and 5098 participants to routine care (on-site sputum smear microscopy with referral-based molecular testing) [52]. In the secondary analysis report of the XPEL-TB 2021 trial, participants were analysed as a subset of the same trial population (n = 5273 Xpert MTB/RIF Ultra; n = 4290 same-day smear microscopy) [53]. The XPEL-TB 2021 trial used the same cluster randomisation in both reports; the secondary analysis did not involve a new randomisation or a separate cohort, but applied different inclusion criteria for the mortality-focused analysis. For Aliasgar 2023, participants were randomised to either same-day Xpert MTB/RIF Ultra (n = 288) or same-day smear microscopy (n = 296) [51]. Neither study assessed our other rapid NAATs of interest.

Outcomes

Critical outcomes

All-cause mortality was evaluated at six months, reported as the unadjusted mortality risk, and at 18 months, reported as the adjusted mortality rate ratio post-TB diagnosis, in the second analysis report of the XPEL-TB 2021 trial [53]. The adjustment accounted for randomisation strata (fixed effect, two levels), the number of individuals treated for TB in the pre-randomisation period, and patient-level covariates such as age, sex, and HIV status [53].

Important outcomes

In the second analysis report of the XPEL-TB 2021 trial, participants lost to follow-up were reported at 18 months post-TB diagnosis [53]. Time to diagnosis was reported by XPEL-TB 2021 in the primary analysis report as an adjusted geometric mean ratio [52]. The XPEL-TB 2021 trial also reported the adjusted geometric mean ratio of the time to treatment [52]. Outcomes were adjusted for cluster-level covariates, including the stratification variable used for randomisation and the number of participants treated for confirmed tuberculosis in the 12-month pre-randomisation period in the XPEL-TB 2021 trial [52]. The arm-specific time (in days) to TB treatment initiation within 60 days of diagnostic testing, with or without culture-based results, was reported by Aliasgar 2023 [51].

Funding sources

Both trials reported receiving external funding [51, 52, 53]. The XPEL-TB 2021 trial was funded by the National Heart, Lung,

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and Blood Institute of the United States National Institutes of Health (NIH) under award numbers R01HL130192 and K12 HL138046. Aliasgar 2023 received funding from the NIH (grant number CRDF-OISE-16-62105), the South African Medical Research Council (grant number RFA-EMU-02-2017), the European and Developing Countries Clinical Trials Partnership (grant numbers TMA-2015SF-1043 and TMA-1051-TESAII), the United Kingdom Medical Research Council (grant number MR/S03563X/1), and the Wellcome Trust (grant number MR/S027777/1). Additionally, Aliasgar 2023 acknowledged individual support from the European & Developing Countries Clinical Trials Partnership (grant number TMA-2015CDF-1052).

We did not contact the authors of the included studies for additional data.

Excluded studies

We excluded 74 studies after screening their full-text reports (see Supplementary material 3). Of these, 24 had an ineligible index test, 23 were ineligible study designs, 13 were ineligible study types, and eight included an ineligible population. Six studies reported on ineligible comparators, and we summarised the characteristics of three of these potentially relevant studies to provide further evidence on the impact of Xpert MTB/RIF Ultra (see Supplementary material 7). Although these three studies were formally excluded due to having ineligible comparators, we summarised their characteristics to provide additional context and insight into the potential impact of Xpert MTB/RIF Ultra, given we included only two trials in the review [56, 57, 58]. The remaining three studies that we excluded due to having ineligible comparators were trial registry records with no publicly available results posted, and therefore we could not summarise them [59, 60, 61].

Risk of bias in included studies

The summary of the risk of bias assessment by outcome is presented in Supplementary material 4. Where outcomes were not reported by the included studies (e.g. no study assessed cure), no risk of bias assessment was possible.

All-cause mortality at 18 months (rate ratio)

For the XPEL-TB 2021 trial, we assessed the risk of bias for all-cause mortality at 18 months as low across all domains. Health centres were randomised at a public ceremony with stratification and restriction to balance groups, and baseline characteristics were comparable across trial arms. Participants and staff were aware of the allocation (on-site versus referral testing), but we judged this unlikely to influence mortality, an objective outcome. Care pathways were delivered as intended, with no cross-over, and analyses were consistent with the intention-to-treat principle. No clusters were lost to follow-up, and 'missingness' was minimal and balanced between arms (86% versus 90%). Mortality status was assessed using registers, phone calls, and home visits, with standardised procedures applied across groups. Although assessors were aware of allocation, we judged this unlikely to influence the objective outcome of mortality. The trial authors applied prespecified verification rules to ascertain vital status, which involved reviewing TB registers, conducting phone calls, and performing home visits. For participants with multiple sources or discrepant dates of death, home-visit data were prioritised, followed by phone calls and registry data. For participants with unknown dates of death, the date was assigned as the midpoint between the last known date of life and the first known date of death. These procedures were planned a priori in the trial protocol. We judged that this approach minimised the risk of performance or detection bias, supporting a low risk of bias for the outcome of all-cause mortality. Mortality was a prespecified important outcome in the trial, and analyses were conducted in accordance with the trial protocol. Overall, we judged this outcome to be at low risk of bias.

All-cause mortality at six months (risk ratio)

For the XPEL-TB 2021 trial, we assessed the risk of bias for all-cause mortality at six months as low across all domains. Randomisation was conducted with stratification and restriction to balance arms, and baseline comparability was demonstrated. Participants and staff were aware of the allocation, but we judged this unlikely to influence mortality, which is an objective outcome. There was no evidence of deviations from the intended interventions, and analyses were conducted according to the intention-to-treat principle. No clusters were lost to follow-up, and missing data were minimal and balanced between groups (86% versus 90%). Mortality was assessed using registers, phone calls, and home visits with standardised procedures. Although assessors were aware of allocation, we judged this unlikely to influence the objective outcome of mortality. The trial authors applied prespecified verification rules, including prioritising home-visit data over phone-call and register data and assigning a midpoint date to unknown deaths, as detailed in the trial protocol. Mortality was a prespecified important outcome in the trial, and analyses were conducted in accordance with the trial protocol. Overall, we judged this outcome to be at low risk of bias.

Loss to follow-up

For the XPEL-TB 2021 trial, we assessed the risk of bias associated with loss to follow-up as low. Health centres were randomised using a robust process, with baseline comparability between arms. As the intervention included visible components (on-site GeneXpert devices, workflow redesign, and monthly report cards), blinding was not feasible; however, no deviations from the intended interventions were reported, and analyses accounted for clustering and stratification. No clusters were lost to follow-up, and although 12% to 15% of participants had missing vital status data, missingness was balanced between arms (P = 0.26). Loss to follow-up was measured consistently across centres using the National Tuberculosis and Leprosy Program (NTLP) registers, with photographs taken every two weeks and centralised abstraction; outcome assessors were aware of allocation, but, in our judgement, measurement was unlikely to be influenced by this knowledge. Handling of loss to follow-up was prespecified, and missingness was transparently reported. Overall, we judged this outcome to be at low risk of bias.

Time to diagnosis (adjusted geometric mean ratio)

For the XPEL-TB 2021 trial, we judged the risk of bias for 'time to diagnosis' to be of concern. Randomisation was robust, with stratification and restriction to balance groups, and baseline characteristics were comparable. Adherence was closely monitored and similar across groups, with no systematic deviations likely to influence outcomes. Missing data were minimal (less than five per cent), reasons for loss to follow-up were unrelated to intervention or outcome, and sensitivity analyses confirmed the robustness of the results. However, insufficient detail was provided on how time to diagnosis was measured in each arm, raising concerns

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about outcome ascertainment. In addition, although the trial was registered, outcome definitions and analysis plans for time-to-event measures were not prespecified in the protocol or registry, raising the possibility of selective reporting. Overall, we judged this outcome as 'some concerns' of risk of bias.

Time to treatment (adjusted geometric mean ratio)

For the XPEL-TB 2021 trial, we judged the risk of bias for 'time to treatment' to be of concern. Randomisation and baseline comparability were appropriate, and adherence was similar across groups with no evidence of deviations from the intended interventions. Missing data were minimal (less than five per cent), and sensitivity analyses supported robustness. However, insufficient detail was provided on how time to initiate treatment was measured in each arm, raising the possibility of differential ascertainment across groups. Furthermore, although the study was registered, outcome definitions and analysis plans for time-to-event outcomes were not prespecified in the protocol or registry, creating a risk of selective reporting. Overall, we judged this outcome as 'some concerns' of risk of bias.

Time to treatment (hazard ratio)

For the Aliasgar 2023 trial, we assessed the risk of bias for 'time to treatment' as low. Randomisation was conducted using block randomisation (block size of four), with baseline comparability across groups. The trial employed an open-label design, but differences in referral pathways were integral to the interventions, and no protocol deviations were reported. Data were available for over 98% of participants. Time to treatment initiation was defined as the date of documented treatment start, an objective and verifiable measure. Although clinical staff were aware of allocation, we judged it unlikely that this knowledge would have biased outcome recording. The outcome was prespecified as the primary endpoint with a 60-day window, and the statistical analysis plan was described a priori in the trial's methods and supplementary materials. Overall, we judged this outcome to be at low risk of bias.

Synthesis of results

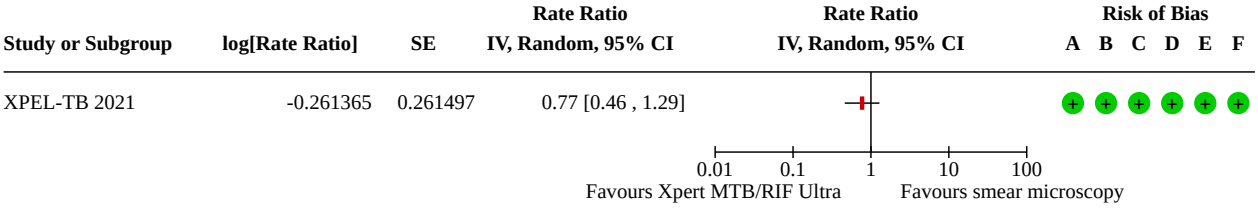
Xpert MTB/RIF Ultra compared to smear microscopy for adults aged ≥ 18 years with presumptive TB

Critical outcomes

All-cause mortality

At 18 months after TB confirmation, XPEL-TB 2021 reported all-cause mortality amongst 8413 participants (see Table 3; Summary of findings 1; Supplementary material 5; Supplementary material 6). The adjusted rate ratio was 0.77 (95% confidence interval (CI) 0.46 to 1.29; 1 RCT, 8413 participants; high-certainty evidence), suggesting a 23% reduction in mortality in the intervention arm using Xpert MTB/RIF Ultra, but the CI included the null value (Figure 2). The unadjusted six-month mortality risk ratio was 0.79, but again, the CI contained the null value (95% CI 0.61 to 1.01; 1 RCT, 8116 participants; high-certainty evidence; Figure 3). We concluded that there is little to no difference in all-cause mortality between Xpert MTB/RIF Ultra and smear microscopy.

Figure 2. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.1: All-cause mortality (measured as a rate ratio) at 18 months

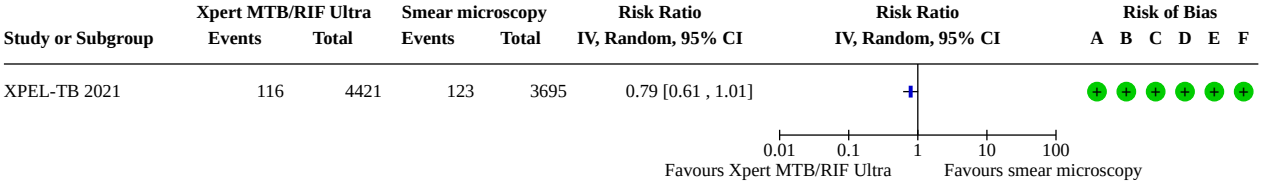


Risk of bias legend

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

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Figure 3. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.2: All-cause mortality (measured as a risk ratio) at six months



Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Cure

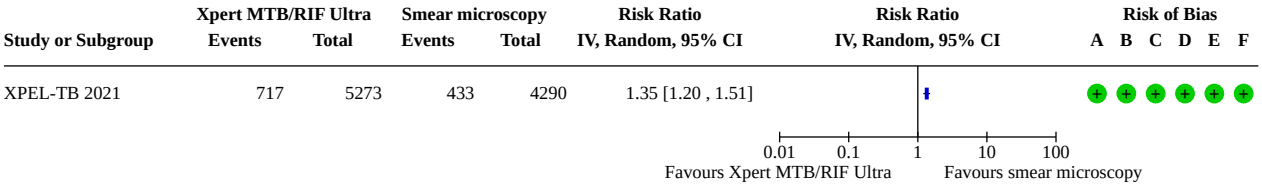
The outcome 'cure' was not reported (Summary of findings 1).

Important outcomes

Proportion of participants lost to follow-up

One trial, which involved 9563 participants, reported loss to follow-up (XPEL-TB 2021). Overall, 1150 participants (12.0%) were lost to follow-up: 717 of 5273 (13.6%) participants in the Xpert MTB/RIF Ultra group and 433 of 4290 (10.1%) participants in the smear microscopy group [53]. The risk ratio indicated that there was a slightly higher risk of loss to follow-up in the Xpert MTB/RIF Ultra arm than in the smear microscopy arm (1.35, 95% CI 1.20 to 1.51; 1 RCT, 9563 participants; high-certainty evidence; Figure 4).

Figure 4. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.3: Loss to follow-up



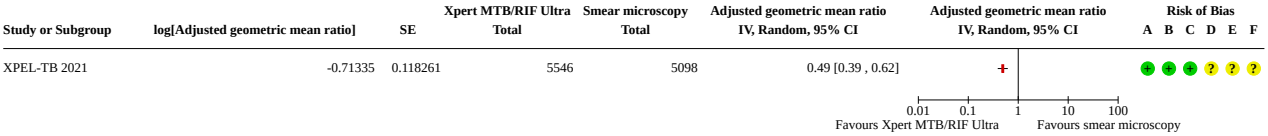
Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Time to diagnosis

XPEL-TB 2021 estimated an adjusted geometric mean ratio of 0.49 (95% CI 0.39 to 0.62; 1 RCT, 10,644 participants; moderate-certainty evidence; Figure 5). This implied that the mean time to TB diagnosis was 51% shorter (95% CI 38 to 61) in the Xpert MTB/RIF Ultra arm than in the smear microscopy arm (Summary of findings 1; Supplementary material 5).

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Figure 5. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.4: Time to diagnosis (measured as adjusted geometric mean ratio)



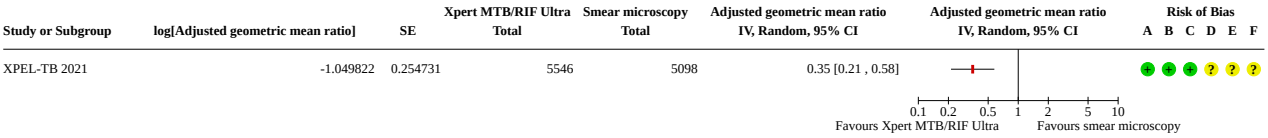
Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Time to treatment

Two studies reported the time to TB treatment initiation, providing moderate-certainty evidence [51, 52].

XPEL-TB 2021 reported time to treatment as an adjusted geometric mean ratio (Figure 6). The adjusted geometric mean ratio indicated that the mean time to TB treatment was 65% shorter for those in the Xpert MTB/RIF Ultra arm than the smear microscopy arm (0.35, 95% CI 0.21 to 0.58; 1 RCT, 10,644 participants; moderate-certainty evidence) (Summary of findings 1; Supplementary material 5).

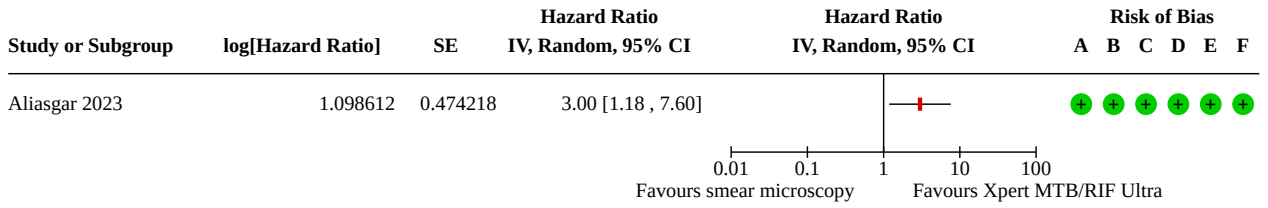
Figure 6. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.5: Time to treatment (measured as adjusted geometric mean ratio)



Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Aliasgar 2023 reported hazard ratios for time to TB treatment initiation (Figure 7). The hazard ratio for time to TB treatment initiation amongst 584 participants, when comparing Xpert MTB/RIF Ultra with smear microscopy without culture-based treatment, was 3.00 (95% CI 1.18 to 7.60; 1 RCT, 584 participants; moderate-certainty evidence) [51].

Figure 7. Forest plot of Xpert MTB/RIF Ultra compared smear microscopy for adults aged ≥18 years with presumptive TB, Outcome 1.6: Time to treatment (measured using hazard ratio)



Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Equity assessment

We did not assess the effect of rapid NAATs across equity-relevant subgroups defined by age, HIV status, drug resistance status, or WHO region, due to the limited number of included studies and data availability. However, the trial by XPEL-TB 2021 included a total of 4190 people living with HIV, while Aliasgar 2023 included 287 people

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living with HIV. No study reported the proportion of participants who were diagnosed with drug-resistant TB.

Reporting biases

We did not assess small-study effects because the number of studies included in each analysis was fewer than 10 [62].

DISCUSSION

Summary of main results

This review aimed to synthesise the effects of LC-aNAATs, LC-mNAATs, and MC-aNAATs on the outcomes of people being investigated for TB. We summarised two RCTs published between 2021 and 2023 that assessed the impact of Xpert MTB/RIF Ultra, a type of LC-aNAAT [51, 52]. Both trials included only adult participants [51, 52]. Data were available for four of our outcomes: all-cause mortality, proportion of participants lost-to-follow-up, time to diagnosis, and time to treatment (see Summary of findings 1).

We found high-certainty evidence that Xpert MTB/RIF Ultra had little to no effect on all-cause mortality at six and 18 months. We also found high-certainty evidence that Xpert MTB/RIF Ultra resulted in a slight increase in the proportion of participants lost to follow-up. Xpert MTB/RIF Ultra likely facilitated faster identification of TB than smear microscopy (moderate-certainty evidence) [51, 52]. Participants diagnosed with TB using Xpert MTB/RIF Ultra were probably quicker to start on treatment than those diagnosed using smear microscopy (moderate-certainty evidence) [51, 52]. We assessed our confidence in the evidence as moderate because the largest trial did not clearly describe its methods for measuring or analysing the time to diagnosis and treatment outcomes.

Neither study measured cure rates.

Limitations of the evidence included in the review

Two trials contributed 11,228 participants to the narrative synthesis of the effects of Xpert MTB/RIF Ultra [51, 52]. Of these, 10,644 participants were from Uganda, a low-income country, while 584 were from South Africa, an upper-middle-income country.

The evidence presented in this review partly addressed our review question. While we aimed to assess the impact of three classes of rapid nucleic acid amplification tests (LC-aNAATs, LC-mNAATs, and MC-aNAATs), the available evidence focused solely on Xpert MTB/RIF Ultra, a type of LC-aNAAT. Data cure rates were unavailable. Therefore, we are unable to assess the effect of LC-aNAATs for improving patient recovery.

Another key limitation is that the included studies only evaluated adults. Although we aimed to consider equity-related issues, such as differences in intervention effects in key subgroups (e.g. children, people living with HIV), this was not possible due to the limited reporting of equity-related data.

We identified different effect measures for the outcomes of time to diagnosis and time to treatment, specifically geometric mean ratios, along with minimal arm-level raw data. The inconsistency in effect measures impeded meaningful comparisons between the two included studies that reported on time to TB treatment initiation.

We assessed our confidence in the evidence identified through the review using the GRADE approach [49]; this is presented in Summary of findings 1. We considered the evidence to be high certainty for all-cause mortality (at both 6 and 18 months) and for the proportion of participants lost to follow-up. The included studies provided moderate certainty of evidence for the impact of Xpert MTB/RIF Ultra on time to diagnosis and time to treatment initiation. We downgraded these two outcomes due to serious concerns about the risk of bias. Neither study reported on cure rates.

Limitations of the review processes

We mitigated limitations in our review process by conducting an extensive literature search, including grey literature sources, with assistance from the CIDG's information specialist and people responding to a WHO public call for data. We mitigated bias by having two review authors independently conduct study selection, data extraction, risk of bias assessments, and GRADE assessments. We did not impose any restrictions on study inclusion based on publication date or language, allowing for a broader range of studies to be considered.

A notable limitation of this review was the identification of three studies with ineligible comparators (Leukes 2024 [56]; Marcy 2023 [57]; Mukoka 2023 [58]). The exclusion of studies with ineligible comparators, including one identified through the WHO public call for data (Leukes 2024), may have limited the breadth of our analysis and potentially affected the representativeness of the included studies. Leukes 2024, which is an ongoing trial in Tanzania and Mozambique evaluating Truenat MTB Plus versus Xpert MTB/RIF Ultra, reported the median time to TB treatment initiation in its preliminary analysis. However, only 70 of the 1980 participants in the comparison arm underwent smear microscopy. Marcy 2023 included paediatric populations in their cluster-RCT evaluating Xpert MTB/RIF Ultra compared to the WHO standard of care for severe pneumonia. Mukoka 2023 performed TB culture tests on the second sputum samples produced by all participants but reported their findings based on the comparison between Xpert MTB/RIF Ultra and chest X-ray in a two-by-two factorial RCT. Therefore, our strict inclusion criteria may have limited our understanding of the comparative effectiveness of diagnostic strategies and their applicability across diverse populations. The requirement for smear microscopy as a comparator also limited the applicability of newer approaches that use non-sputum samples (e.g. stool testing for children, for which Xpert MTB/RIF Ultra was recommended), for which smear microscopy would not have been performed.

We did not include adverse events as a specific outcome, although this is recommended in the Cochrane MECIR standards [36]. The omission of this outcome limited the review's comprehensiveness. However, the tests we assessed are in vitro diagnostics, which we do not think pose direct harm to patients.

Another limitation of this review is that early design choices, such as excluding observational studies without concurrent controls, may have influenced the comprehensiveness of our findings. These decisions were made to enhance internal validity by focusing on studies with a lower risk of bias. However, this approach may have excluded valuable real-world data and other potentially relevant test types.

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Agreements and disagreements with other studies or reviews

Our search identified a systematic review and meta-analysis by Lee 2022 [63], which included 45 studies: 26 focused on drug-susceptible tuberculosis (DS-TB) and 20 on drug-resistant tuberculosis (DR-TB) [64]. The authors aimed to evaluate the impact of rapid molecular tests on diagnostic and treatment delays in TB. Their eligibility criteria permitted inclusion of both the standard Xpert MTB/RIF and Xpert MTB/RIF Ultra tests. However, it was not clearly stated whether any studies specifically involving Xpert MTB/RIF Ultra were identified and included in their final analyses. Similar to our findings, their analyses found that the use of Xpert likely reduced diagnostic and treatment delays compared with sputum microscopy, which could inform public health strategies for TB control. However, when discussing the implications of Lee's 2022 findings, it is crucial to note the lack of clarity regarding the inclusion of Xpert MTB/RIF Ultra, which may bias the results toward standard Xpert MTB/RIF. This limitation could explain why our review identified only three studies that assessed the impact of Xpert MTB/RIF Ultra. Our review identified studies on Xpert MTB/RIF Ultra exclusively from Africa, whereas Lee 2022 included studies from multiple low- and middle-income countries (LMICs), with most conducted in Africa. Consistent with our comparator of interest, Lee 2022 also compared rapid NAATs with smear microscopy of sputum samples.

Unlike Lee 2022, we did not find studies reporting on DS-TB or DR-TB. The applicability of our results may be limited to individuals without TB drug resistance, which could restrict their generalisability to real-world settings. Additionally, the evidence presented in our review may overlook critical differences in how rapid molecular tests affect diagnosis and treatment outcomes, potentially leading to biased conclusions about the overall impact of rapid NAATs. While we aimed to include people with presumptive TB irrespective of their age and for both PTB and EPTB, Lee 2022 focused on adult populations only with presumed cases of PTB. Lee 2022 did not evaluate the impact of rapid NAATs on all-cause mortality, cure, or the proportion of participants that were lost to follow-up.

The findings of our review align with those of the 2021 Cochrane review on Xpert MTB/RIF, which assessed the impact of the earlier version of the test on patient outcomes [29]. Consistent with our review, the 2021 review found no conclusive evidence that Xpert MTB/RIF reduced all-cause mortality or improved overall treatment success, despite its greater diagnostic accuracy than smear microscopy. However, it reported a probable reduction in mortality among people living with HIV and a higher proportion of bacteriologically confirmed cases starting treatment, findings that are consistent with our review's conclusion that Xpert MTB/RIF Ultra likely reduces delays in diagnosis and treatment initiation. Although Xpert MTB/RIF Ultra clearly improves case detection compared with earlier tests, the evidence on its effect on long-term patient-important outcomes, such as cure, is lacking [52]. This underscores the need for further research to determine whether advances in diagnostic accuracy translate into meaningful health benefits, particularly across diverse patient populations and sample types.

AUTHORS' CONCLUSIONS

Implications for practice

Using Xpert MTB/RIF Ultra tests to diagnose tuberculosis (TB) showed little to no difference in all-cause mortality at six and 18 months compared with smear microscopy. Neither study included in this review reported cure rates. The limited data prevented a complete assessment of the effects of rapid nucleic acid amplification tests (NAATs) on patient health. While evidence from the studies suggested that Xpert MTB/RIF Ultra probably reduced the time taken until diagnosis and time taken until initiation of TB treatment, healthcare practitioners should interpret these findings with caution, given the limited data. The use of Xpert MTB/RIF Ultra is associated with a slight increase in the risk of loss to follow-up compared with smear microscopy.

Equity-related implications for practice

Equity-based assessments could not be conducted due to insufficient data.

Implications for research

The limited data from randomised controlled trials (RCTs) on critical outcomes, such as all-cause mortality, and the complete lack of data from RCTs for critical outcomes like cure rate, revealed a gap in the evidence base regarding the real-world impact of rapid NAATs such as Xpert MTB/RIF Ultra. Although we rated the certainty of evidence for all-cause mortality as high, this reflects confidence in the available estimate rather than the breadth or generalisability of the evidence. Future research should prioritise both all-cause mortality and cure rate across the three classes of rapid NAATs to provide a comprehensive understanding of how these diagnostic tools influence patient survival and long-term treatment success. Studies included in this review were conducted within the World Health Organization (WHO) African region. Evidence from other WHO regions with high TB incidence, such as the South-East Asia and Western Pacific regions, will help generalise findings on the impact of low-complexity automated NAATs [3].

In addition, future research is needed to explore the performance of Xpert MTB/RIF Ultra in people with drug-susceptible TB and drug-resistant TB, which were not examined in the studies included in this review. Understanding its effectiveness across diverse drug-resistance profiles would be crucial for broader application. Studies should aim for methodological rigour and transparent reporting of study methods.

Equity-related implications for research

We suggest that future studies be designed with equity considerations in mind to better understand the effects of rapid nucleic acid amplification tests across subpopulations.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD016194.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

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Supplementary material 4 Risk of bias

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Additional evidence from relevant RCTs excluded due to ineligible comparators

ADDITIONAL INFORMATION

Acknowledgements

We acknowledge the WHO Guideline Development Group (GDG).

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article.

- Sign-off Editor(s): Geraint Rhys Davies, University of Liverpool (first decision); Toby Lasserson, Acting Editor-in-Chief (final decision)
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Joanne Duffield, Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Lisa Wydrzynski, Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Mary Gaeddert, Heidelberg University (clinical/content review); CM Zuraidah (patient and public review); Jo-Ana Chase, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review). Two additional peer reviewers provided clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

Jenifer Akoth Otieno (JAO): study selection; data extraction; risk of bias assessment; data cleaning and analyses; data interpretation; contribution to the summary of findings tables and certainty of the evidence assessment; writing the first draft; and contribution to subsequent revisions of the review.

Lisa Malesi Were (LMW): study selection; data extraction; risk of bias assessment; data cleaning; and multiple revisions of the review.

Victoria Lutje (VL): literature search for relevant studies; contribution to writing the search methods section of the first draft; and multiple revisions of the review.

Katie Scandrett (KS): review conceptualisation; data analyses; data interpretation; and multiple revisions of the review.

Yemisi Takwoingi (YT): review conceptualisation; coordination of the review process; data analyses; data interpretation; and multiple revisions of the review.

Eleanor Ochodo (EO): arbitration of disagreements regarding study selection; data extraction; and risk of bias assessments; review conceptualisation; coordination of the review process; contribution to the summary of findings tables and certainty of the evidence assessment; and multiple revisions of the review.

Declarations of interest

JAO has no known conflict of interest.

LMW has no known conflict of interest.

VL is an Information Specialist for the Cochrane Central Editorial Service, and was an Information Specialist for Cochrane Infectious Diseases until the group closed on 31 December 2024. She was not involved in the editorial process for this review.

KS has no known conflict of interest.

YT is a member of the Cochrane Editorial Board and was an Editor with Cochrane Infectious Diseases until the group closed on 31 December 2024. She was not involved in the editorial process for this review.

EO declares payments made directly to her by the World Health Organisation (WHO) for conducting this review, including travel support related to this work. She was an Editor with Cochrane Infectious Diseases until the group closed on 31 December 2024. She was not involved in the editorial process for this review.

Of note, none of the eligible studies were authored by any of the review authors.

Sources of support

Internal sources

- Kenya Medical Research Institute, Kenya

  Jenifer Otieno, Lisa Were, and Eleanor Ochodo received internal institutional support in the form of salary and access to resources.

External sources

- World Health Organization, Other

  The WHO provided external support through funding for literature searches, study selection, data extraction, travel, and publication costs. The funder had no involvement in the development of the protocol or in the conduct of the review. The views and opinions expressed therein are those of the review authors and do not necessarily reflect those of the funder.

Registration and protocol

Cochrane approved the proposal for this review in June 2024.

Protocol (2023) OSF | Generic impact SR protocol for mWRDs Final.pdf.

Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database (see Supplementary material 1), characteristics of inldued studies (see Supplementary material 2), characterisitcs of excluded studies (see Supplementary material 3), risk of bias (see Supplementary material 4), analyses (see Supplementary material 5), data package (see Supplementary

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material 6), and additional evidence from relevant RCTs excluded due to ineligble comparators (see Supplementary material 7). Analyses and data management were conducted in Cochrane's authoring tool, RevMan, using the built-in computational methods [65]. The template data extraction form from Covidence is available from the authors on reasonable request [44].

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REFERENCES

1. Bloom BR, Atun R, Cohen T, Dye C, Fraser H, Gomez GB, et al. Chapter 11: Tuberculosis. In: Holmes KK, Bertozzi S, Bloom BR, Jha P, editor(s). Major Infectious Diseases. Third edition. Washington (DC): The International Bank for Reconstruction and Development/The World Bank, 2017:233-4. [DOI: 10.1596/978-1-4648-0524-0] [URL: https://www.dcp-3.org/sites/default/files/chapters/DCP3%20Major%20Infectious%20Diseases_Ch11.pdf]

2. World Health Organisation. Global Tuberculosis Report 2022; 2022. Available at https://iris.who.int/bitstream/handle/10665/363752/9789240061729-eng.pdf?sequence=1.

3. World Health Organisation. Global tuberculosis report 2024; 2024. Available at https://iris.who.int/bitstream/handle/10665/379339/9789240101531-eng.pdf?sequence=1.

4. Sharma SK, Mohan A. Extrapulmonary tuberculosis. *Indian Journal of Medical Research* 2004;**120**(4):316-53. [PMID: 15520485]

5. Marais BJ, Gie RP, Schaaf HS, Hesseling AC, Obihara CC, Starke JJ, et al. The natural history of childhood intra-thoracic tuberculosis: a critical review of literature from the pre-chemotherapy era. *International Journal of Tuberculosis and Lung Disease* 2004;**8**(4):392-402. [PMID: 15141729]

6. Marais BJ. Childhood tuberculosis: epidemiology and natural history of disease. *Indian Journal of Pediatrics* 2011;**78**(3):321-7. [DOI: 10.1007/s12098-010-0353-1]

7. Jenkins HE, Yuen CN, Rodriguez CA, Nathavitharana RR, McLaughlin MM, Donald P, et al. Mortality in children diagnosed with tuberculosis: a systematic review and meta-analysis. *Lancet. Infectious Diseases* 2016;**17**(3):285-95. [DOI: 10.1016/S1473-3099(16)30474-1]

8. Litvinjenko S, Magwood O, Wu S, Wei X. Burden of tuberculosis among vulnerable populations worldwide: an overview of systematic reviews. *Lancet. Infectious Diseases* 2023;**23**(12):1395–407. [DOI: 10.1016/S1473-3099(23)00372-9]

9. World Health Organisation. 2018-2022. Status update: reaching the targets in the political declaration of the United Nations General Assembly high-level meeting on the fight against tuberculosis, September 2023; 15 September 2023. Available at https://www.who.int/publications/m/item/WHO-UCN-TB-2023.4.

10. World Health Organisation. Meeting report of the WHO expert consultation on the definition of extensively drug-resistant tuberculosis: 27-29 October 2020; 22 January 2021. Available at https://www.who.int/publications/i/item/9789240018662.

11. World Health Organisation. WHO consolidated guidelines on tuberculosis. Module 3: diagnosis–rapid diagnostics for tuberculosis detection, third edition; 20 March 2024. Available at https://www.who.int/publications/i/item/9789240089488.

12. World Health Organisation. WHO operational handbook on tuberculosis. Module 3: diagnosis-rapid diagnostics for tuberculosis detention; 2021. Available at https://iris.who.int/handle/10665/342369.

13. World Health Organization. Extension: Public call for data on data to inform WHO guidelines development process. WHO Website January 2024. [URL: https://www.who.int/news-room/articles-detail/extension-public-call-for-data-on-data-to-inform-who-guidelines-development-process]

14. Theron G, Zijenah L, Chanda D, Clowes P, Rachow A, Lesosky, et al. Feasibility, accuracy, and clinical effect of point-of-care Xpert MTB/RIF testing for tuberculosis in primary-care settings in Africa: a multicentre, randomised, controlled trial. *Lancet* 2014;**383**(9915):424-35. [DOI: 10.1016/s0140-6736(13)62073-5]

15. Zifodya JS, Kreniske JS, Schiller I, Kohli M, Dendukuri N, Schumacher SG, et al. Xpert Ultra versus Xpert MTB/RIF for pulmonary tuberculosis and rifampicin resistance in adults with presumptive pulmonary tuberculosis. *Cochrane Database of Systematic Reviews* 2021, Issue 2. Art. No: CD009593. [DOI: 10.1002/14651858.CD009593.pub5]

16. Penn-Nicholson A, Gomathi SN, Ugarte-Gil C, Meaza A, Lavu E, Patel P, et al. A prospective multicentre diagnostic accuracy study for the Truenat tuberculosis assays. *European Respiratory Journal* 2021;**58**(5):2100526. [DOI: 10.1183/13993003.00526-2021]

17. Nikam C, Jagannath M, Narayanan MM, Ramanabhiraman NV, Kazi M, Shetty A, et al. Rapid diagnosis of Mycobacterium tuberculosis with Truenat MTB: a near-care approach. *PloS One* 2013;**8**(1):e51121. [DOI: 10.1371/journal.pone.0051121]

18. Nikam C, Kazi M, Nair C, Jaggannath M, Manoj M, Vinaya R, et al. Evaluation of the Indian Truenat micro RT-PCR device with GeneXpert for case detection of pulmonary tuberculosis. *International Journal of Mycobacteriology* 2014;**3**(3):205-10. [DOI: 10.1016/j.ijmyco.2014.04.003]

19. Bruno L, Janne A, Terhi M, Silja M, Anu P. Evaluation of the STANDARD M10 MDR-TB and MTB/NTM assays for the detection of *Mycobacterium tuberculosis*, rifampicin and isoniazid resistance, and non-tuberculous mycobacteria in a low-incidence setting. *Journal of Clinical Microbiology* 2024;**62**(10):e0040224. [DOI: 10.1128/jcm.00402-24]

20. Georghiou SB, De Vos M, Velen K, Miotto P, Colman RE, Cirillo DM, et al. Designing molecular diagnostics for current tuberculosis drug regimens. *Emerging Microbes and Infections* 2023;**12**(1):2178243. [DOI: 10.1080/22221751.2023.2178243]

21. Inbaraj LR, Daniel J, Narayanan MKS, Srinivasalu VA, Bhaskar A, Rajendran P, et al. TB-LAMP (loop-mediated isothermal amplification) for diagnosing pulmonary tuberculosis in children. *Cochrane Database of Systematic Reviews* 2023, Issue 9. Art. No: CD015806. [DOI: 10.1002/14651858.CD015806]

22. Dayal R, Yadav A, Agarwal D, Kumar M, Kamal R, Singh D, et al. Comparison of diagnostic yield of tuberculosis loop-

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mediated isothermal amplification assay with cartridge-based nucleic acid amplification test, acid-fast bacilli microscopy, and mycobacteria growth indicator tube culture in children with pulmonary tuberculosis. *Journal of the Pediatric Infectious Diseases Society* 2021;**10**(2):83-7. [DOI: 10.1093/jpids/piaa019]

23. Fan L, Guan B, Cheng M, Liu C, Tian Y, Li R, et al. A comprehensive evaluation of a loop-mediated isothermal amplification assay for the diagnosis of pulmonary tuberculosis in children using bronchoalveolar lavage fluid. *Infection and Drug Resistance* 2022;**15**:975-87. [DOI: 10.2147/IDR.S354660]

24. Tomotada I, Toshiaki S, Kozaburo H. Loop-mediated isothermal amplification for direct detection of *Mycobacterium tuberculosis* complex, *M. avium*, and *M. intracellulare* in sputum samples. *Journal of Clinical Microbiology* 2003;**41**(6):2616-22. [DOI: 10.1128/JCM.41.6.2616-2622.2003]

25. Deepali J, Manoj N, Mandira V, Daman S. Evaluation of improved IS6110 LAMP assay for diagnosis of pulmonary and extra pulmonary tuberculosis. *Journal of Microbiological Methods* 2017;**139**:87-91. [DOI: 10.1016/j.mimet.2017.05.007]

26. Nandal L, Perumal R, Naidoo K. Rapid molecular assays for the diagnosis of drug-resistant tuberculosis. *Infection and Drug Resistance* 2022;**15**:4971-84. [DOI: 10.2147/IDR.S381643]

27. Chien J-Y, Chiu W-H, Chien S-T, Chiang C-J, Yu C-J, Hsueh P-R. Mutations in gyrA and gyrB among fluoroquinolone- and multidrug-resistant *Mycobacterium tuberculosis* isolates. *Antimicrobial Agents and Chemotherapy* 2016;**60**(4):2090-6. [DOI: 10.1128/AAC.01049-15]

28. Conkle-Gutierrez D, Kim C, Ramirez-Busby SM, Modlin SJ, Mansjö M, Werngren J, et al. Distribution of common and rare genetic markers of second-line-injectable-drug resistance in *Mycobacterium tuberculosis* revealed by a genome-wide association study. *Antimicrobial Agents and Chemotherapy* 2022;**66**(6):e0207521. [DOI: 10.1128/aac.02075-21]

29. Haraka F, Kakolwa M, Schumacher SG, Nathavitharana RR, Denkinger CM, Gagneux S, et al. Impact of the diagnostic test Xpert MTB/RIF on patient outcomes for tuberculosis. *Cochrane Database of Systematic Reviews* 2021, Issue 5. Art. No: CD012972. [DOI: 10.1002/14651858.CD012972.pub2]

30. MacPherson P, Houben RM, Glynn JR, Corbett EL, Kranzer K. Pre-treatment loss to follow-up in tuberculosis patients in low- and lower-middle-income countries and high-burden countries: a systematic review and meta-analysis. *Bulletin of the World Health Organization* 2014;**92**(2):126-38. [DOI: 10.2471/blt.13.124800]

31. World Health Organisation. WHO consolidated guidelines on tuberculosis. Module 3: diagnosis: rapid diagnostics for tuberculosis detection, 2021 update; 7 July 2021. Available at https://www.who.int/publications/i/item/9789240029415.

32. WHO Global Tuberculosis Programme. Public notice: Guideline Development Group meeting on nucleic acid amplification tests (NAATs) for detection of TB and DR-TB; 6–10 May 2024. Geneva: World Health Organization. Website April 2024. [URL: https://www.who.int/publications/m/item/public-notice--guideline-development-group-meeting-on-nucleic-acid-amplification-tests-(naats)-for-detection-of-tb-and-dr-tb]

33. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: 10.1136/bmj.n71]

34. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from https://www.cochrane.org/handbook. [URL: www.training.cochrane.org/handbook]

35. Pitcher N, Mitchell D, Hughes C. Template and guidance for writing a Cochrane Plain language summary. Version 1 January 2022. Available at https://training.cochrane.org/handbook/current/guidance-writing-cochrane-plain-language-summary.pdf.

36. Higgins J, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological Expectations of Cochrane Intervention Reviews (MECIR): Standards for the conduct of new Cochrane Intervention Reviews, and the planning and conduct of updates. Cochrane, 2023. Available at https://community.cochrane.org/mecir-manual.

37. Takwoingi Y, Ochodo E, Otieno, Steingart K, Scandrett K, Were L, et al. Impact of rapid nucleic acid amplification tests for tuberculosis on patient outcomes: systematic review (generic protocol). Open Science Framework 20231-20. [URL: https://osf.io/puax2]

38. Singh JA, Siddiqi M, Parameshwar P, Chandra-Mouli V. World Health Organization guidance on ethical considerations in planning and reviewing research studies on sexual and reproductive health in adolescents. *Journal of Adolescent Health* 2019;**64**(4):427-9. [DOI: 10.1016/j.jadohealth.2019.01.008]

39. Campelo TA, De Sousa PRC, De Lima Nogueira L, Frota CC, Antas PRZ. Revisiting the methods for detecting *Mycobacterium tuberculosis*: what has the new millennium brought thus far? *Access Microbiology* 2021;**3**(8):000245. [DOI: 10.1099/acmi.0.000245]

40. Sethi S, Sharma S, Dhatwalia SK, Sharma S, Aggarwal AN, Yadav R. Comparing diagnostic accuracy: Xpert MTB/RIF Ultra vs Xpert MTB/RIF for diagnosis of pulmonary tuberculosis. Indian Journal of Tuberculosis (in press). [DOI: 10.1016/j.ijtb.2024.04.009]

41. World Health Organisation. Meeting report of the WHO expert consultation on drug-resistant tuberculosis treatment outcome definitions: 17-19 November 2020; 7 April 2021. Available at https://www.who.int/publications/i/item/9789240022195.

42. STOP TB Partnership. New diagnostic working group. website 2025. [URL: https://www.stoptb.org/who-we-are/stop-tb-working-groups/new-diagnostics]

43. A global alliance for diagnostics. Website. [URL: https://www.finddx.org/]

Copyright © 2025 The Authors. Cochrane Database of Systematic Reviews published by John Wiley & Sons, Ltd. on behalf of The Cochrane Collaboration.

44. Covidence. Version accessed 19 October 2023. Melbourne, Australia: Veritas Health Innovation, 2023. Available at https://covidence.org.

45. Page MJ, Higgins JP, Sterne JA. Chapter 13: Assessing risk of bias due to missing evidence in a meta-analysis. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available at https://www.cochrane.org/handbook.

46. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898. [DOI: 10.1136/bmj.l4898]

47. Eldridge S, Campbell M, Campbell M, Dahota A, Giraudeau B, Higgins H, et al. Revised Cochrane risk of bias tool for randomized trials (RoB 2). Additional considerations for cluster-randomized trials (RoB 2 CRT); 18 March 2021. Available at https://www.riskofbias.info/welcome/rob-2-0-tool/rob-2-for-cluster-randomized-trials. [URL: https://www.riskofbias.info/welcome/rob-2-0-tool/rob-2-for-cluster-randomized-trials]

48. Higgins JP, Li T, Sterne JA; on behalf of the RoB 2 working group on crossover trials. Revised Cochrane risk of bias tool for randomized trials (RoB 2). Additional considerations for crossover trials; 18 March 2021. Available at https://sites.google.com/site/riskofbiastool/welcome/rob-2-0-tool/rob-2-for-crossover-trials. [URL: https://www.riskofbias.info/welcome/rob-2-0-tool/rob-2-for-crossover-trials]

49. Schünemann H, Brożek J, Guyatt G, Oxman A, editor(s). Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach (updated October 2013). GRADE Working Group, 2013. Available from https://gdt.gradepro.org/app/handbook/handbook.html.

50. Balshem H, Helfand M, Schünemann HJ, Oxman AD, Kunz R, Brozek J, et al. GRADE guidelines: 3. Rating the quality of evidence. *Journal of Clinical Epidemiology* 2011;**64**(4):401-6. [DOI: 10.1016/j.jclinepi.2010.07.015]

51. Aliasgar E, Philippa R, Suzette O, Michele T, Anil P, Richard M, et al. Comparison of two diagnostic intervention packages for community-based active case finding for tuberculosis: an open-label randomized controlled trial. *Nature Medicine* 2023;**29**:1009-16. [DOI: 10.1038/s41591-023-02247-1]

52. Cattamanchi A, Tania FR, Talemwa N, Katherine A, Mariam N, Denis O, et al. Multicomponent strategy with decentralized molecular testing for tuberculosis. *New England Journal of Medicine* 2021;**385**(26):2441-50. [DOI: 10.1056/NEJMoa2105470] [PMID: 34936740]

53. Katamba A, Mochizuki T, Nalugwa T, Nantale M, Oyuku D, Nabwire S, et al. Impact of a multicomponent strategy including decentralized molecular testing for tuberculosis on mortality: planned analysis of a cluster-randomized trial in Uganda. *eClinical Medicine* 2024;**78**:102953. [DOI: 10.1016/j.eclinm.2024.102953]

54. NCT03044158. GeneXpert performance evaluation for linkage to tuberculosis care (XPEL-TB). https://clinicaltrials.gov/study/NCT03044158 (first posted 6 February 2017).

55. NCT03168945. Xpert active case-finding trial 2: community-based active case-finding for tuberculosis in South Africa (XACT-2). https://clinicaltrials.gov/study/NCT03168945 (first posted 30 May 2017).

56. Leukes VN, Hella J, Sabi I, Cossa M, Khosa C, Erkosar B, et al. Study protocol: a pragmatic, cluster-randomized controlled trial to evaluate the effect of implementation of the Truenat platform/MTB assays at primary health care clinics in Mozambique and Tanzania (TB-CAPT CORE). *BMC Infectious Diseasses* 2024;**24**(1):107. [DOI: 10.1186/s12879-023-08876-8]

57. Marcy O, Wobudeya E, Font H, Vessiere A, Chabala C, Khosa C, et al. Effect of systematic tuberculosis detection on mortality in young children with severe pneumonia in countries with high incidence of tuberculosis: a stepped-wedge cluster-randomised trial. *Lancet. Infectious Diseases* 2023;**23**(3):341-51. [DOI: 10.1016/S1473-3099(22)00668-5]

58. Mukoka M, Twabi HH, Msefula C, Semphere R, Ndhlovu G, Lipenga T, et al. Utility of Xpert MTB/RIF Ultra and digital chest radiography for the diagnosis and treatment of TB in people living with HIV: a randomised controlled trial (XACT-TB). *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2023;**117**(1):28-37. [DOI: 10.1093/trstmh/trac079]

59. ISRCTN13936377. Truenat molecular testing for the effective diagnosis of tuberculosis [Localised mobile active case finding with Truenat molecular testing for the effective diagnosis of tuberculosis]. https://trialsearch.who.int/Trial2.aspx?TrialID=ISRCTN13936377 (first posted 13 February 2023).

60. PACTR202403493915530. Platform assessing regimens and durations in a global multisite consortium for TB [PARADIGM4TB (UNITE4TB – 01)] [A seamless phase 2B/2C platform trial to evaluate multiple regimens and durations of treatment in pulmonary tuberculosis]. https://trialsearch.who.int/Trial2.aspx?TrialID=PACTR202403493915530 (first posted 9 February 2024).

61. PACTR202410708311054. Conditional cash transfers and pre- and post-TB test counselling to reduce total loss to follow up across the continuum of care. https://trialsearch.who.int/Trial2.aspx?TrialID=PACTR202410708311054 (first posted 25 September 2024).

62. Nguyen P, Kanukula R, McKenzie J, Alqaidoom Z, Brennan S, Haddaway N, et al. Changing patterns in reporting and sharing of review data in systematic reviews with meta-analysis of the effects of interventions: cross sectional meta-research study. *BMJ* 2022;**379**:e072428. [DOI: 10.1136/bmj-2022-072428]

63. Lee JH, Garg T, Lee J, McGrath S, Rosman L, Schumacher SG, et al. Impact of molecular diagnostic tests on diagnostic and treatment delays in tuberculosis: a systematic review and meta-analysis. *BMC Infectious Diseases* 2022;**22**(1):940. [DOI: 10.1186/s12879-022-07855-9]

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64. Lee JH, Garg T, Lee J, McGrath S, Rosman L, Schumacher SG, et al. Impact of molecular diagnostic tests on diagnostic and treatment delays in tuberculosis: a systematic review and meta-analysis. *BMC Infectious Diseases* 2022;**22**(1):940.

65. Review Manager (RevMan). Version 8.6.1. The Cochrane Collaboration, 2024. Available at https://revman.cochrane.org.

ADDITIONAL TABLES

Table 1. Treatment outcome definitions for both drug-resistant and drug-susceptible tuberculosis

Patient outcome	Definition
Treatment failed	A patient whose treatment regimen needed to be terminated or permanently changed[a] a to a new regimen or treatment strategy
Cure	A pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who completed treatment as recommended by the national policy, with evidence of bacteriological response[b] and no evidence of failure
Treatment completed	A patient who completed treatment as recommended by the national policy, whose outcome does not meet the definition for cure or treatment failure
Died	A patient who died[c] before starting a treatment or during treatment
Lost-to-follow-up	A patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more
Not evaluated	A patient for whom no treatment outcome was assigned[d]
Treatment success	The sum of cured and treatment completed
Time to diagnosis	The duration between a patient's first reported TB-related symptom onset and the formally confirmed detection of TB or diagnosis
Time to treatment	The period between the diagnosis of TB and the initiation of treatment

Adapted from the WHO expert consultation meeting report on drug-resistant tuberculosis treatment outcome definitions [41]

[a]Reasons for the change include:

- no clinical response and/or no bacteriological response (see note 'b');
- adverse drug reactions; or
- evidence of additional drug resistance to medicines in the regimen.

[b]"Bacteriological response" refers to bacteriological conversion with no reversion.

- "Bacteriological conversion" describes a situation in a patient with bacteriologically confirmed TB where at least two consecutive cultures (for DR-TB and DS-TB) or smears (for DS-TB only), taken on different occasions at least 7 days apart, are negative.
- "Bacteriological reversion" describes a situation where at least two consecutive cultures (for DR-TB and DS-TB) or smears (for DS-TB only), taken on different occasions at least 7 days apart, are positive either after the bacteriological conversion or in patients without bacteriological confirmation of TB.

[c]Patient died for any reason

[d]This includes cases "transferred out" to another treatment unit and those whose treatment outcome is unknown; however, it excludes those lost to follow-up.

Abbreviations

DR-TB: drug-resistant TB; **DS-TB:** drug-susceptible TB; **TB:** tuberculosis

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Table 2. Key characteristics of included studies

Study identification	Study design	Country (WHO region)	Facility (geographical area)	Target population	Index test (sample type)	Comparator test (sample type)	Outcomes	Funding
XPEL-TB 2021[a] [52, 53]	Pragmatic parallel cRCT with 2 arms (multi-centre)	Uganda (African region)	20 health centres (urban & rural areas)	**In the primary analysis report:** 10,644 adults ≥ 18 years, whose TB records had been entered into treatment registers irrespective of their HIV status **In the secondary analysis report:** 9563 adults ≥ 18 years, whose TB records had been entered into treatment registers with known HIV status	Xpert MTB/RIF Ultra (sputum)	Smear microscopy (sputum)	**In the primary analysis report:** Critical outcome: number of patients treated[b] Important outcomes: mean number of days taken to diagnose and start treatment for TB **In the secondary analysis report:** Critical outcome: all-cause mortality rate censored at 18 months Important outcomes: 6-month mortality risk, proportion of participants lost to follow-up at 18 months	NHLBI part of the US NIH under award numbers R01HL130192 and K12 HL138046
Aliasgar 2023 [51]	Parallel iRCT with 2 arms (single-centre)	South Africa (African region)	Community-based scalable mobile clinic (per-urban area)	584 adults > 18 years, at least one TB symptom for 2 weeks or PLHIV irrespective of symptoms	Xpert MTB/RIF Ultra (sputum)	Smear microscopy (TB culture was also done on the second sputum sample)	Critical outcome: arm-specific time to treatment initiation within 60 days of diagnostic testing Important outcomes: not reported	NIH (grant number CRDF-OISE-16-62105), the SAMRC (grant number RFA-EMU-02-2017), the EDCTP (grant numbers TMA-2015SF-1043, TMA-1051-TESAII and TMA-2015CDF-1052), the UKMRC (grant number MR/S03563X/1), and the Wellcome Trust (grant number MR/S027777/1)

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[a]There were two separate analysis reports for the XPEL-TB trial, with each focusing on different outcomes.

[b]Outcome not relevant to our review

Abbreviations

cRCT: cluster-randomised controlled trial; **EDCTP:** European & Developing Countries Clinical Trials Partnership; **HIV:** human immunodeficiency virus; **iRCT:** individual randomised controlled trial; **MTB:** *Mycobaterium tuberculosis*; **NHLBI:** National Heart, Lung, and Blood Institute; **NIH:** National Institutes of Health; **PLHIV:** people living with HIV; **RIF:** resistance to rifampin; **SAMRC:** South African Medical Research Council; **TB:** tuberculosis; **UKMRC:** United Kingdom Medical Research Council; **US:** United States; **WHO:** World Health Organization

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Table 3. Descriptive summary of results from the included studies

Study identification (study design)	Type of analysis & sample size	Lost-to-follow-up (n/N, %)			Participant outcome results	
		Intervention arm	Comparison arm	Overall (at study endpoint)	Critical outcomes	Important outcomes
XPEL-TB 2021(cRCT) [52, 53]	ITT N = 10,644 I = 5546 C = 5098 In the secondary analysis report: complete case analysis N = 9563 I = 5273 C = 4290	717/5273 (13.6%)	433/4290 (10.1%)	1150/9563 (12.0%)	**Mortality rate (censored at 18 months):** the adjusted rate ratio (aRR) was 0.77 (95% CI 0.47 to 1.28), comparing the Xpert MTB/RIF Ultra arm (145 deaths/3655 person-years) with the smear microscopy arm (154 deaths/3015 person-years)	**6-month mortality risk:** the overall unadjusted 6-month mortality risk in the Xpert MTB/RIF Ultra arm was 2.5% (116 deaths among 4421 individuals) compared with 3.1% (123 deaths among 3695 individuals) in the smear microscopy arm. **Time to TB diagnosis:** adjusted geometric mean ratio (days): 51% (95% CI 38 to 61) lower in the intervention group **Time to TB treatment:** adjusted geometric mean ratio (days): 65% (95% CI 44 to 79) lower in the intervention group
Aliasgar 2023 (iRCT) [51]	Modified ITT N = 584 I = 288 C = 296	Not reported	Not reported	Not reported	**Time to TB treatment, days:** same-day treatment (excluding culture-based treatment initiation): HR: 3.0 (95% CI 1.2 to 7.7), P = 0.02	Not reported

C: control; **CI:** confidence interval; **HR:** hazard ratio; **I:** intervention; **iRCT:** individually randomised controlled trial; **ITT:** intention-to-treat analysis; **N:** number; **P:** probability; **TB:** tuberculosis

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